

Here is one example of an output from `model.show_topics()`:

```
[(0, '0.304*bass" + 0.200*tuba" + 0.182*deep" + 0.165*horn" + 0.113*pike" +  
0.036*catapult'), (1, '0.331*pike" + 0.209*horn" + 0.201*catapult" + 0.121*bass" +  
0.120*deep" + 0.018*tuba'), (2, '0.340*pike" + 0.194*bass" + 0.176*deep" + 0.167*horn"  
+ 0.091*catapult" + 0.033*tuba")]
```

Each tuple in the list corresponds to a topic (there are three possible topics), with the first element in the tuple (either 0, 1, or 2) representing the topic number. The tuple represents the probability distribution for that topic amongst the six words of our vocabulary—the probability associated with each word for that topic is written next to the word. Here are the inferred beta vectors from the above output side by side with the actual beta vector from the topic-word matrix that was used to train the model:

Topic 0 From Model Output

Output: [0.304 (bass), 0.200 (tuba), 0.182 (deep), 0.165 (horn), 0.113 (pike), 0.036 (catapult)]

True (topic 2 / 3rd row in beta matrix matches most closely with this topic):

[0.0 (pike), 0.3 (bass), 0.2 (deep), 0.2 (horn), 0.0 (catapult), 0.3 (tuba)]

Topic 1 From Model Output

Output: [0.331 (pike), 0.209 (horn), 0.201 (catapult), 0.121 (bass), 0.120 (deep), 0.018 (tuba)]

True (topic 0 / 1st row in beta matrix matches most closely with this topic):

[0.3 (pike), 0.3 (horn), 0.3 (catapult), 0.0 (bass), 0.1 (deep), 0.0 (tuba)]

Topic 2 From Model Output

Output: [0.340 (pike), 0.194 (bass), 0.176 (deep), 0.167 (horn), 0.091 (catapult), 0.033 (tuba)]

True (topic 1 / 2nd row in beta matrix matches most closely with this topic):

[0.4 (bass), 0.0 (tuba), 0.2 (deep), 0.0 (horn), 0.4 (pike), 0.0 (catapult)]

The inferred beta vectors don't seem very accurate because there is a lot of deviation / noise when compared to the true vectors. The probabilities within each topic amongst the words, are generally closer to each other when looking at the inferred beta vectors as compared to the actual beta vectors (the model predicts less variation within the topics than there actually is). Since the model did not map the topics (0, 1, and 2) in the same way that we did initially, I Just looked at the outputs to see which probabilities most closely aligned when matching up the topics. Having more training data, for example by generating more texts for our corpus in order to train the LDA model, could help make the inferred beta vectors more accurate.